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To cite this article: Phuc Khang Nguyen and Quang Linh Huynh 2025 *J. Phys.: Conf. Ser.* **2949** 012004

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A Machine Learning Approach in EEG-Based Assessment of Cognitive Load by Mental Arithmetic Tasks

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Abstract. In this study, we investigate the characteristics of electroencephalogram (EEG) signals during mental arithmetic tasks. The publicly available dataset used in this study includes EEG recordings from 36 healthy volunteers performing serial subtraction tasks, with data collected using a 23-channel Neurocom EEG system. The study aims to explore the changes in brain activity during cognitive workload by analyzing the power spectral density, entropy, and statistical features of the EEG signals. Key features such as band power in delta, theta, alpha, beta, and gamma frequency bands, entropy, mean power, standard deviation, kurtosis, skewness, autocorrelation, and RMS energy were extracted from each channel. The subjects were categorized into good and bad counters based on their arithmetic performance. The extracted features were then used to train machine learning models, including Linear Discriminant Analysis (LDA), k-Nearest Neighbors (KNN), and Support Vector Machine (SVM), to classify the cognitive workload levels. The KNN model demonstrated the highest overall performance with an F1 score of 0.857. This research highlights the potential of using EEG features to assess cognitive workload and provides a foundation for future studies on brain-computer interfaces and cognitive neuroscience.

1. Introduction

Electroencephalography (EEG) has long been a valuable tool for investigating brain activity in various cognitive tasks, particularly in mental arithmetic, which serves as a standard model for cognitive workload studies. EEG records electrical activity from the brain using electrodes placed on the scalp and provides insights into different frequency bands that correspond to specific cognitive and emotional processes [1]. Mental arithmetic tasks, which involve solving mathematical problems without external aids, are commonly used to induce cognitive load in experimental settings [2].

The growing interest in understanding the relationship between EEG signals and cognitive workload has led to significant advances in neuroimaging techniques and machine learning approaches. Mental arithmetic, often perceived as stressful, requires sustained attention and working memory engagement, which can alter brainwave patterns across multiple frequency

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bands. Studies have demonstrated that cognitive workload primarily affects the theta (4–8 Hz) and alpha (8–13 Hz) bands, with increased activity in the beta (13–30 Hz) and gamma (>30 Hz) bands during high cognitive demand [3], [4].

Previous research has focused on the power spectral density (PSD) of EEG signals, entropy, and statistical features as indicators of mental workload. By extracting key features such as band power, entropy, mean power, and kurtosis from the EEG signals, machine learning algorithms can effectively classify different levels of cognitive load [5], [6]. This study aims to further explore these characteristics by analyzing EEG signals collected during mental arithmetic tasks performed by healthy volunteers.

This study builds on previous research that uses serial subtraction tasks to induce cognitive load, allowing us to examine how the brain responds to mental workload through EEG signals. The key contributions of this research lie in both the data processing and analysis pipeline as well as the machine learning models employed. We developed a comprehensive process to handle raw EEG data, applying techniques such as Independent Component Analysis (ICA) for artifact removal and extracting critical features like band power, entropy, mean power, and standard deviation. These features provide valuable insights into how brain activity changes with varying cognitive load. Additionally, we evaluated machine learning models—including Linear Discriminant Analysis (LDA), k-Nearest Neighbors (KNN), and Support Vector Machine (SVM)—to classify different levels of cognitive load. Among these, KNN demonstrated the highest accuracy and F1 score, making it particularly effective for real-time cognitive load assessment. The findings not only contribute to a better understanding of brain dynamics under stress but also have practical applications in developing more accurate brain-computer interface (BCI) systems, which could enhance real-time cognitive monitoring in education and healthcare settings.

2. Material and Methods

2.1 Materials

The EEG dataset used in this study was obtained from 36 healthy volunteers aged 18 to 26 years (M_{age} = 18.6 years, SD = 0.87 years) [7]. All participants were right-handed and had normal or corrected-to-normal vision, with no history of neurological or psychiatric disorders. The participants were students from the Educational and Scientific Centre "Institute of Biology and Medicine," National Taras Shevchenko University of Kyiv, Ukraine. The participants provided informed consent prior to the experiment, and the study was conducted in accordance with the Declaration of Helsinki [8].

2.2 EEG Recording

EEG signals were recorded using a 23-channel Neurocom EEG system following the international 10/20 electrode placement system. The electrodes were positioned over frontal (Fp1, Fp2, F3, F4), central (C3, Cz, C4), parietal (P3, Pz, P4), occipital (O1, O2), and temporal (T3, T4) regions. A sampling rate of 500 Hz was employed, with a high-pass filter of 0.5 Hz, a low-pass filter of 45 Hz, and a notch filter of 50 Hz to remove electrical interference. The inter-electrode impedance was kept below 5 kΩ throughout the recording.

2.3 Experimental Task and Data Processing

The participants were asked to perform a serial subtraction task, a commonly used method to induce cognitive load. The task required the participants to subtract a two-digit number (e.g., 17) from a randomly chosen four-digit number (e.g., 4753) repeatedly over a 4-minute period.

This task was performed following a 3-minute resting period, during which participants were instructed to close their eyes and remain relaxed.

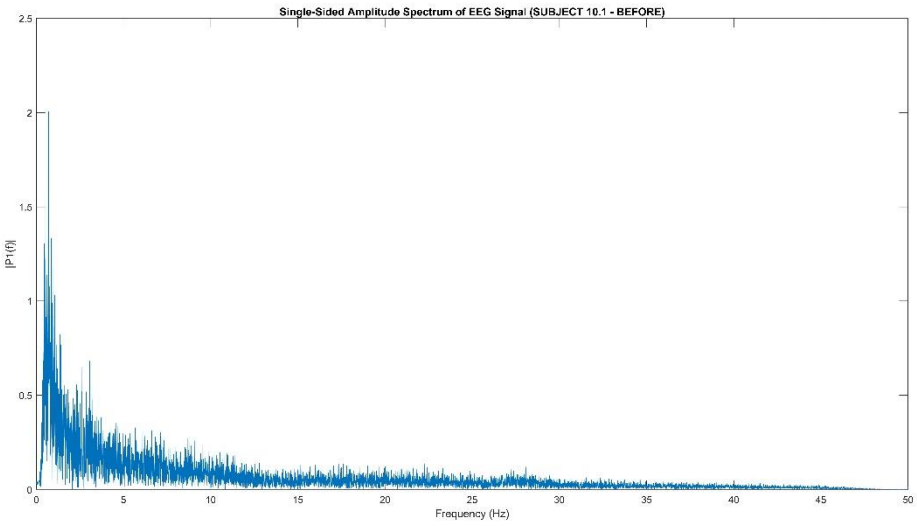
Figure 1 summarizes the data for the participating subjects. Females are denoted by "F", whereas males are denoted by "M". The "number of subtractions" is defined as the difference between the starting 4-digit number and the subtraction result, divided by the subtrahend. It is important to note that if the computation for a certain subject is incorrect, the resulting value is not an integer. We determined the "count quality" ranking based on the number of subtractions for groups "B" and "G" [7].

Name	Age	Gender	Number of Subtractions	Count Quality
Subject0	21	F	9.7	B
Subject1	18	F	29.35	G
Subject2	19	F	12.88	G
Subject3	17	F	31	G
Subject4	17	F	8.6	B
Subject5	16	F	20.71	G
Subject6	18	M	4.35	B
Subject7	18	F	13.38	G
Subject8	26	M	18.24	G
Subject9	16	F	7	B
Subject10	17	F	1	B
Subject11	18	F	26	G
Subject12	17	F	26.36	G
Subject13	24	M	34	G
Subject14	17	F	9	B
Subject15	17	F	22.18	G
Subject16	17	F	11.59	G
Subject17	17	F	28.7	G
Subject18	17	F	20	G
Subject19	22	M	7.06	B
Subject20	17	F	15.41	G
Subject22	19	F	4.47	B
Subject21	20	F	1	B
Subject23	16	F	27.47	G
Subject24	17	M	14.76	G
Subject25	17	M	30.53	G
Subject26	17	F	13.59	G
Subject27	19	F	34.59	G
Subject28	19	F	27	G
Subject29	19	M	16.59	G
Subject30	17	M	10	B
Subject31	19	F	19.88	G
Subject32	20	F	13	G
Subject33	17	M	21.47	G
Subject34	18	F	31	G
Subject35	17	F	12.18	G

Figure 1. *Subjects and population data [7]*

Raw EEG data was preprocessed to remove artifacts caused by eye movements, muscle activity, and cardiac pulsations. Independent Component Analysis (ICA) was applied to eliminate these artifacts. Additionally, a 30 Hz high-pass filter and a 50 Hz notch filter were used to further clean the data. Each subject's EEG data was segmented into 60-second intervals, and only data from the arithmetic task were included in the analysis.

After removing artifacts and preprocessing the EEG data, Fourier analysis was performed to identify the frequency spectrum of the signals under different conditions. The following figures illustrate the Fourier analysis for subject 10 in two scenarios: during rest and while performing the arithmetic task.



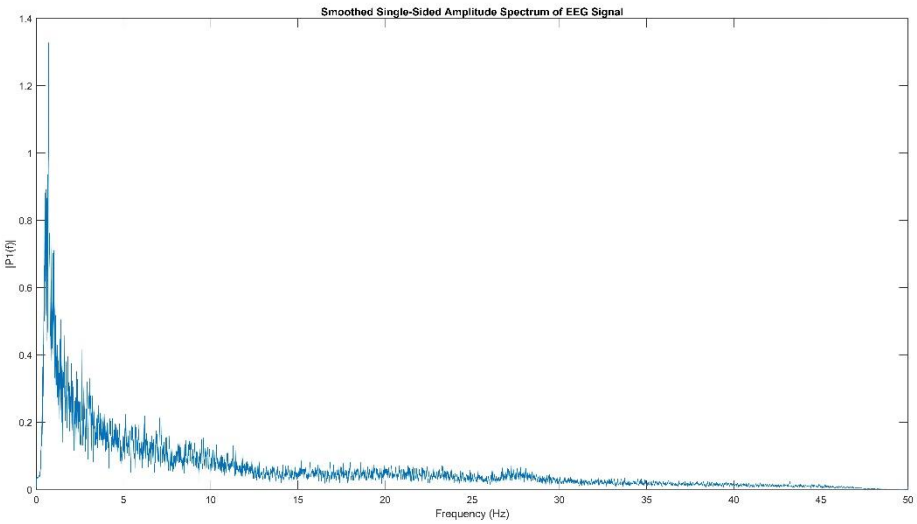
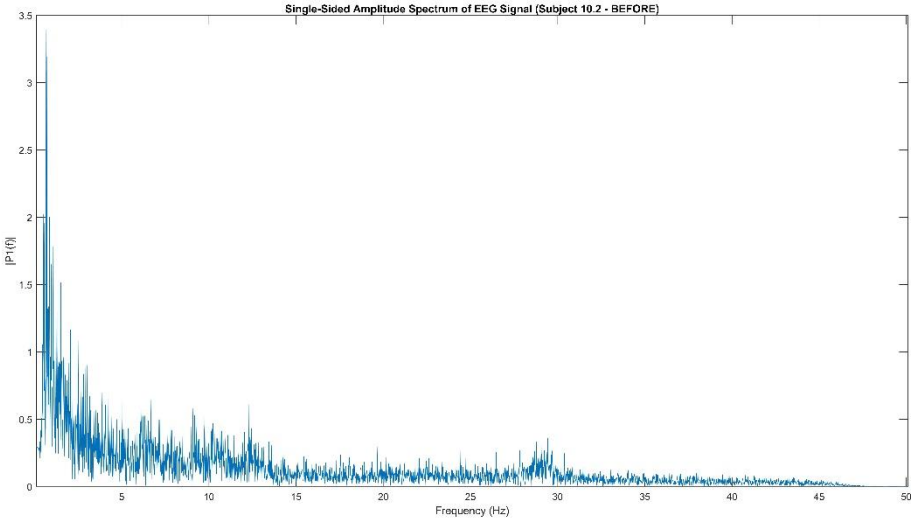


Figure 2 and Figure 3. FFT analysis of subject 10 during rest.



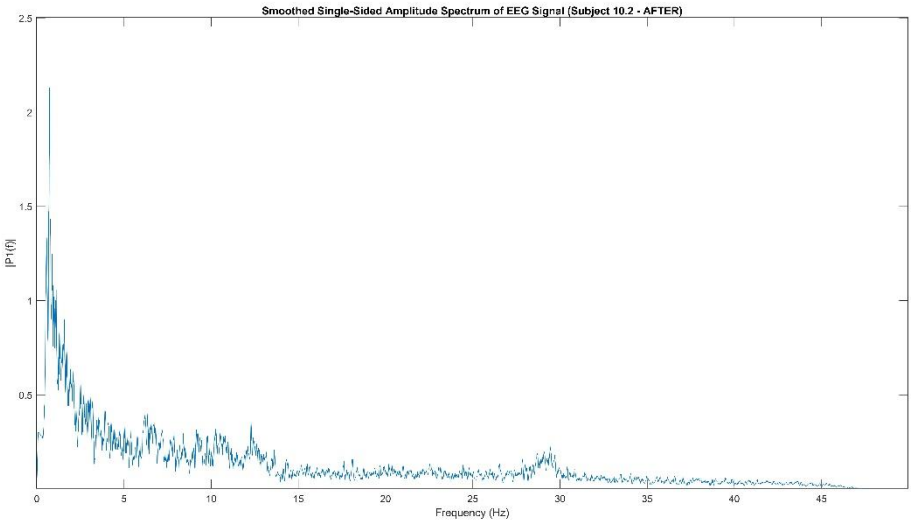


Figure 3 and Figure 4. FFT analysis of subject 10 during arithmetic task

In the rest state, there are clear peaks in the low frequency range, especially in the 8-13 Hz range, indicating the presence of alpha waves, which are characteristic of a relaxed state. After smoothing, the alpha frequency peaks (8-13 Hz) are still clear, along with some activity at lower frequencies, possibly theta (4-8 Hz) or delta (0.5-4 Hz) waves.

During the computational task, we see a clear decrease in alpha waves and an increase in activity at higher frequencies, especially in the beta range (13-30 Hz), indicating increased cognitive activity and concentration. After smoothing, higher frequency activities are still evident, especially beta waves and possibly some gamma waves (>30 Hz).

2.4 Feature Extraction

In this study, we extracted several key features from the EEG signals to assess cognitive load during mental arithmetic tasks. These features include band power, entropy, mean power, standard deviation, kurtosis, skewness, autocorrelation, and RMS energy. Each of these features plays a

crucial role in analyzing and classifying different levels of cognitive load, especially when utilizing machine learning models to evaluate brain activity.

Table 1. Description of Extracted Features Used in EEG-Based Cognitive Load Classification Research [9]

Feature Name	Description	Extraction Domain
Band Power	Total power of the EEG signal within specific frequency bands, representing specific cognitive states.	Frequency domain
Entropy	Measures the complexity and unpredictability of the EEG signal.	Time domain
Mean Power	Average power of the EEG signal, reflecting the level of brain activity over a specific period of time.	Frequency domain
Standard Deviation	Measures the variability of the EEG signal, indicating stability or change in cognitive states.	Time domain
Kurtosis	Assesses the "tailedness" or concentration of the EEG signal distribution, detecting subtle abnormalities.	Statistical domain
Skewness	Measures the asymmetry of the EEG signal distribution, useful in analyzing slight distortions in the signal.	Statistical domain
Autocorrelation	Measures the similarity between the EEG signal and its lagged version, detecting repetitive patterns.	Time domain

RMS Energy	Total RMS energy of the EEG signal, providing information on the overall activation level of the brain.	Frequency domain

[a] Hojat Ghimatgar, Kamran Kazemi, Mohammad Sadegh Helfroush, Ardalan Aarabi, "An automatic single-channel EEG-based sleep stage scoring method based on hidden Markov Model," Journal of Neuroscience Methods, vol. 324, 2019, article 10832, pp.4

2.4.1. Band power

Band power is calculated by summing the squared values of the EEG signal within a specific frequency band. The frequency bands are categorized as follows: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (>30 Hz).

$$P_{band} = \sum_{f1}^{f2} |\mathrm{X}(f)|^2 \tag{1}$$

where X(f) represents the Fourier-transformed values of the EEG signal at frequency f, f_1 and f_2 are the limits of the frequency band. Band power is a critical metric for understanding brain activity across different cognitive states. Each frequency band corresponds to specific brain states, such as relaxation or intense cognitive effort . For example, delta and theta waves are often associated with relaxation or sleep, while beta and gamma waves are linked to higher cognitive functions and focus. Analyzing band power allows us to determine the level of focus and stress experienced by participants during complex arithmetic tasks, providing essential insights for classifying cognitive load levels. This analysis is particularly valuable in developing machine learning models that assess cognitive load based on EEG signals.

Previous studies have focused on the relationship between band power and various cognitive states. Menon et al. (2000) demonstrated that increased beta activity during arithmetic tasks is indicative of heightened cognitive load, while decreased alpha activity corresponds to higher stress levels [3]. Similarly, Pinheiro-Chagas et al. (2018) found that reduced alpha waves corresponded with increased cognitive load during tasks that required significant concentration [4].

2.4.2. Entropy

Entropy is calculated using the Shannon entropy formula:

$$H(X) = -\sum_i p(x_i) \log\big(p(x_i)\big) \tag{2}$$

where $p(x_i)$ is the probability of the occurrence of value x_i in the EEG signal.

Entropy measures the complexity and unpredictability of the EEG signal. Higher entropy values often reflect irregular and chaotic signals, which can be associated with increased cognitive load or stress. In this study, entropy helps capture the complexity of brain activity under different cognitive states. Using entropy is an effective approach to identifying chaotic patterns in EEG signals, thereby improving the classification of cognitive load levels using machine learning models [10].

Entropy has been widely applied in EEG studies to assess cognitive and emotional states. Peng et al. (1995) showed that entropy can reflect the complexity of EEG signals across different cognitive states, with higher entropy values often linked to increased stress and cognitive load [6]. Soleymani et al. (2012) used entropy to analyze changes in EEG signals related to emotional responses, demonstrating that entropy is a powerful tool for emotion recognition based on EEG [5].

2.4.3. Mean Power

Mean power is calculated by averaging the squared values of the EEG signal:

$$Mean\ power = \frac{1}{N}\sum_{i=1}^{N} {x_i}^2 \tag{3}$$

where x_i is the value of the EEG signal at time *i*, and *N* is the total number of samples.

Mean power is an important metric for evaluating the overall activity of the brain over a specific period. It allows us to compare the level of activation between different cognitive states, such as rest versus task performance. In this study, mean power helps determine the level of brain activity and contributes to classifying different levels of cognitive load. Studies have shown that mean power increases in higher frequency bands, such as beta and gamma, during tasks that require intense focus.

Finlay and colleagues (2016) explored changes in mean power across different psychological states and found that mean power increases in the beta band during tasks that demand high cognitive effort [11]. Zyma et al. (2019) also found that differences in mean power between resting and task states reflect brain activation levels, indicating that mean power is a crucial metric for evaluating brain activity in this study [12].

2.4.4. Standard Deviation

Standard deviation is calculated using the following formula:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2} \tag{4}$$

where x_i is the value of the EEG signal at time iii, μ is the mean value, and *N* is the total number of samples.

Standard deviation measures the variability of brain activity, helping to identify stability or change in the EEG signal. A high standard deviation may indicate instability or rapid shifts in cognitive states, which is particularly important when analyzing different stages of arithmetic tasks. This feature enhances the accuracy of classifying cognitive load levels in the study.

Standard deviation is commonly used in EEG research to assess signal variability under different experimental conditions. Menon et al. (2000) showed that increased standard deviation corresponds to more complex task phases, reflecting significant changes in brain activity [3]. Soleymani et al. (2012) also noted that standard deviation is a key metric in distinguishing cognitive and emotional states, especially during tasks requiring high cognitive effort [5].

2.4.5. Kurtosis and Skewness

Kurtosis is calculated as follows:

$$\mathrm{K} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \mu}{\sigma}\right)^4 \tag{5}$$

where μ is the mean value and σ is the standard deviation.

Skewness is calculated as follows:

$$S = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \mu}{\sigma}\right)^3 \tag{6}$$

Kurtosis and skewness are important statistical measures that describe the shape of the EEG signal distribution. Kurtosis assesses the "tailedness" or concentration of the distribution, while skewness measures its asymmetry. These features are particularly useful in detecting subtle abnormalities in the EEG signal related to different cognitive states. For example, high kurtosis may indicate extreme values, reflecting intense cognitive activity. The use of kurtosis and skewness in the study improves the classification of cognitive load levels with machine learning models.

Höll and Kantz (2015) utilized kurtosis and skewness to evaluate EEG signal variability in studies of circadian rhythms, highlighting that these metrics can reflect significant changes in brain activity [13]. Soleymani et al. (2012) applied kurtosis and skewness in EEG analysis for emotion recognition, demonstrating that these metrics are powerful tools for analyzing signal characteristics related to cognitive load [5].

2.4.6. Autocorrelation

Autocorrelation is calculated using the following formula:

$$R(k) = \frac{\sum_{i=1}^{N-k}(x_i . x_{i+k})^3}{\sum_{i=1}^{N}(x_i)^2} \tag{7}$$

where k is the time lag, and x_i is the value of the signal at time i.

Autocorrelation measures the degree of similarity between the EEG signal and its lagged version, helping to detect repetitive patterns in the signal, such as circadian rhythms or prolonged cognitive activities. This is particularly important when analyzing EEG signals related to arithmetic tasks, where repetitive patterns may reflect continuous cognitive load or cyclical brain activity. Using autocorrelation in this study enhances the detection and classification of cognitive states.

Pinheiro-Chagas et al. (2018) applied autocorrelation to analyze repetitive patterns in EEG signals, showing that autocorrelation can reveal biological rhythms in brain activity during complex cognitive tasks [4]. Autocorrelation has been shown to be an effective tool in analyzing cyclic EEG signals, particularly in studies of cognitive states and stress.

2.4.7. RMS Energy

RMS energy is calculated as follows:

$$E = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i)^2} \tag{8}$$

where x_i is the value of the EEG signal at time i.

RMS energy is a critical metric for assessing the overall energy of the EEG signal, especially in contexts where signal amplitude varies significantly. RMS energy provides information about the overall activation level of the brain, particularly during tasks requiring high focus and problem-solving. The use of RMS energy in this study helps to more accurately assess the cognitive load of participants during mental arithmetic tasks, supporting the classification and prediction of different cognitive states with machine learning models.

Peng et al. (1995) demonstrated that RMS energy is an effective metric for evaluating the overall energy changes in EEG signals across different cognitive states [6]. RMS energy has been

widely used in the analysis of biological signals, particularly in EEG studies related to cognitive and emotional activities.

3. Classification

3.1. Machine Learning Models

To classify the cognitive workload based on the extracted EEG features, we employed several machine learning models, including Linear Discriminant Analysis (LDA), k-Nearest Neighbors (KNN), and Support Vector Machine (SVM). These models are chosen for their proven effectiveness in handling high-dimensional data and their ability to generalize well across different types of EEG datasets.

Linear Discriminant Analysis (LDA) is a linear classification model that is particularly effective when the features are linearly separable. It reduces the dimensionality of the feature space while maintaining the class separability, making it a suitable choice for EEG-based classification tasks [14].

k-Nearest Neighbors (KNN) is a non-parametric method that classifies instances based on the majority vote of their neighbors. This method is advantageous in scenarios where the decision boundary is complex and non-linear. KNN has been shown to perform well in EEG classification tasks, especially when combined with feature selection techniques [15].

Support Vector Machine (SVM) is a powerful model that works by finding the hyperplane that best separates the different classes in the feature space. SVM is particularly useful in handling high-dimensional data and has been widely used in EEG signal classification due to its robustness and ability to manage non-linear decision boundaries through the use of kernel functions [16].

The combination of these machine learning models allows for a comprehensive evaluation of the EEG data, providing insights into the most effective methods for classifying cognitive workload. The results obtained from these models will be analyzed to determine the most suitable approach for real-time cognitive load assessment.

These findings are consistent with similar studies in the field of cognitive load assessment. For instance, research by *Wu and Zhang (2006)* demonstrated that KNN performs well in EEG classification tasks due to its ability to handle complex decision boundaries effectively, making it a strong candidate for applications involving high-dimensional EEG data [15]. While models like SVM have shown strong performance in general EEG classifications, particularly in managing non-linear separations as highlighted by *Cortes and Vapnik (1995)* [16], KNN's adaptability and capacity to capture intricate patterns within EEG data often allow it to achieve comparable or even superior accuracy under conditions with well-preprocessed data. Additionally, studies such as those by *Blankertz et al. (2008)* have emphasized that although SVM is effective, especially in brain-computer interface (BCI) applications where precision is critical, KNN's performance in this study suggests that it can offer a robust alternative with high classification accuracy when feature selection and preprocessing are appropriately managed [14]. Furthermore, compared to Linear Discriminant Analysis (LDA), which generally works well for linearly separable data, KNN's ability to adapt to complex, non-linear data makes it particularly advantageous for EEG-based cognitive load assessments. These comparisons further validate the choice of KNN in this study as a highly effective model for cognitive load classification based on EEG signals.

4. Result and Discussion

4.1 Result

The analysis of EEG signals during mental arithmetic tasks across all subjects and channels revealed significant variations in key metrics, providing valuable insights into how cognitive load affects brain activity. The band power values demonstrated expected trends under high cognitive load conditions. Specifically, delta power (0.5–4 Hz) averaged 105.67 μV^2, with a noticeable decrease during tasks requiring higher cognitive engagement, indicating reduced relaxation. Theta power (4–8 Hz), with a mean value of 16.83 μV^2, also decreased as cognitive load increased, reflecting a shift from relaxation to task-related focus. Alpha power (8–13 Hz) averaged 17.68 μV^2 and decreased during mental tasks, consistent with findings that lower alpha activity is associated with higher cognitive demands. In contrast, beta power (13–30 Hz), which had a mean value of 11.94 μV^2, increased during tasks requiring sustained attention and working memory, indicating active mental processing. Gamma power (>30 Hz), with a mean value of 2.59 μV^2, also showed an increase, further corroborating its association with higher cognitive load and complex brain functions.

Entropy, a measure of the complexity of neural activity, exhibited high values across all subjects, indicating the chaotic nature of EEG signals under cognitive load. Entropy values ranged from 9261.02 μV^2 to 10901.85 μV^2, exemplified by Subject 22-2, which had one of the highest entropy values, reflecting the subject's high cognitive load during the tasks. Mean power across all channels and subjects provided further evidence of increased brain activity during cognitive tasks, with an overall mean power of 136.52 μV^2, where higher values indicated greater brain activation in response to the mental arithmetic tasks.

Conversely, kurtosis and skewness, which describe the shape of the EEG signal distribution, were less influential in differentiating cognitive load levels. Kurtosis values ranged from 2.81 (subject 30-2) to 8.01 (subject 22-2), indicating varying signal extremities. Skewness ranged from -1.01 (subject 25-2) to 0.73 (subject 32-2), reflecting different levels of signal asymmetry among subjects. While these features were found to be less impactful in this specific context, they were still included in the analysis due to their theoretical significance. Kurtosis and skewness are often used in EEG studies to provide insights into the distribution of signal amplitudes, which can indicate the presence of outliers or asymmetric patterns that might be relevant under different conditions or tasks. Their initial inclusion was based on the premise that such characteristics might reveal subtle differences in brain activity patterns under varying cognitive loads, even if they ultimately contributed less to the classification performance in this study.

The results underscore that band power and entropy are the most reliable indicators of cognitive load, with significant changes observed in the beta and gamma frequency bands, alongside consistently high entropy values during mental arithmetic tasks.

Given these observations, the EEG features were utilized in machine learning models to classify cognitive load levels. The models—Linear Discriminant Analysis (LDA), k-Nearest Neighbors (KNN), and Support Vector Machine (SVM)—were evaluated for their classification accuracy. The KNN model achieved the highest performance, with an F1 score of 0.857, outperforming both LDA and KNN models. This suggests that KNN, with its kernel-based approach, is particularly well-suited for distinguishing between different cognitive load states based on EEG features.

Table 2. Performance metrics of the LDA, SVM, and KNN models in classifying cognitive load levels based on EEG features.

Model	Accuracy	Precision	Recall	F1 score
LDA	0.714	0.75	0.9	0.818
KNN	**0.786**	**0.818**	**0.9**	**0.857**
SVM	0.714	0.75	0.9	0.818

4.2. Discussion and Future works

The observed variations in EEG signals across different frequency bands provide valuable insights into how the brain responds to increasing cognitive load. Specifically, the increased activity in the beta (13–30 Hz) and gamma (>30 Hz) bands during complex tasks suggests that these frequencies are closely associated with processes involving attention and working memory [3, 5]. These findings align with the global workspace theory, which posits that higher cognitive functions involve widespread neural activation across the cortex [1]. The significant decrease in delta (0.5–4 Hz) and theta (4–8 Hz) band power further underscores the shift from relaxed states to task-oriented mental engagement, reflecting the brain's dynamic adaptation to cognitive demands.

The analysis also highlighted the critical role of entropy as a measure of the complexity of neural activity. High entropy values across subjects indicated that increased cognitive load correlates with more chaotic and disordered neural patterns, aligning with theories that suggest cognitive effort leads to increased signal complexity. This insight is particularly valuable for distinguishing between different levels of cognitive demand, reinforcing the utility of entropy in cognitive load assessment.

The superior performance of the KNN model in classifying cognitive load levels underscores its robustness in handling high-dimensional EEG data. The kernel-based approach of KNN allows it to capture complex, non-linear patterns in the data, which is crucial when dealing with the intricate relationships between EEG features. The findings suggest that KNN, when combined with carefully selected features such as band power and entropy, is an effective tool for cognitive load assessment. The feature selection process was pivotal in enhancing the model's performance, demonstrating that focusing on features directly linked to cognitive processes can significantly improve classification accuracy. For instance, band power, particularly in the beta and gamma frequencies, was shown to be a strong indicator of cognitive load, while entropy provided insights into the complexity of neural activity under stress.

The study is not without limitations. The relatively small sample size may restrict the generalizability of the results, as EEG data can be highly individualistic. Additionally, the study primarily focused on a limited number of EEG features, and there may be other relevant features that were not considered. Future studies should include larger and more diverse populations to validate these findings across different demographics. Moreover, exploring advanced machine learning models, such as deep learning approaches, could potentially yield even higher accuracy and better generalization in cognitive load assessment tools based on EEG signals.

Building on the insights gained from this study, future research should aim to address the limitations mentioned above. Increasing the sample size and diversity of participants will be crucial in ensuring that the findings are robust and applicable to a broader population.

Futhermore, expanding the scope of feature extraction to include more nuanced EEG characteristics, such as phase-amplitude coupling or cross-frequency interactions, could provide a more comprehensive understanding of how cognitive load manifests in neural activity.

Exploring deep learning techniques, such as convolutional neural networks (CNNs) or recurrent neural networks (RNNs), may also offer significant improvements in classifying cognitive load levels. These models are particularly well-suited for capturing complex temporal and spatial patterns in EEG data, which could lead to more accurate and generalizable cognitive load assessment tools. Furthermore, integrating multimodal data sources, such as combining EEG with physiological signals like heart rate variability or skin conductance, could enhance the robustness of cognitive load predictions.

Finally, there is considerable potential to develop real-time applications of cognitive load assessment using EEG signals. Such applications could be used in adaptive learning environments, where the system adjusts the difficulty of tasks based on the user's cognitive load, or in workplace settings to monitor and optimize employee performance under varying cognitive demands. The integration of these advanced techniques and broader datasets will pave the way for more sophisticated and practical tools for assessing cognitive load in real-world scenarios.

5. Conclusion

This study aimed to enhance our understanding of how cognitive load impacts brain activity, specifically through the analysis of EEG signals during mental arithmetic tasks. By focusing on key EEG features such as beta and gamma band power and entropy, we successfully identified reliable indicators of cognitive load. The KNN model, which outperformed other machine learning models, demonstrated its effectiveness in classifying cognitive load levels, reinforcing the potential of these features in practical applications.

Our findings align with the broader goals outlined in the introduction, emphasizing the importance of accurate cognitive load assessment in fields such as adaptive learning and workplace efficiency. While the results are promising, particularly in the identification of high-dimensional EEG patterns, the study's limitations—such as sample size—highlight the need for further research. Future work should build on these insights, exploring more advanced machine learning techniques and expanding the scope of EEG feature extraction to include additional neural characteristics.

In conclusion, this research contributes valuable knowledge to the field of cognitive neuroscience and lays the groundwork for future developments in real-time cognitive load assessment tools, potentially transforming how cognitive states are monitored and managed in various real-world applications.

Acknowledgment

The authors are grateful to Khai Le Quoc for their contributions and advice at the beginning of this study. We acknowledge Ho Chi Minh City University of Technology (HCMUT), VNU-HCM for supporting this study.

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